
Kafka Components

1. Introduction

Apache Kafka is a distributed event streaming platform used for building real-time data pipelines and event-driven applications.

Kafka works using multiple core components that together enable message production, storage, and consumption.

2. Kafka Components Overview

The main components of Kafka are:

- Producer
- Consumer
- Topic
- Broker
- Partition
- Consumer Group
- Zookeeper (legacy / optional in newer Kafka versions)

Each component plays an important role in the data flow.

3. Producer

A **Producer** is an application that sends data (messages/events) to Kafka.

Responsibilities:

- Creates messages
- Sends messages to Kafka topics
- Publishes events when something happens in the system

Example:

Order Service sends an event like:
"Order Created" → Kafka Topic

4. Consumer

A **Consumer** is an application that reads messages from Kafka topics.

Responsibilities:

- Reads messages from Kafka
- Processes data
- Performs business actions like updating database or triggering workflows

Example:

Inventory Service consumes "Order Created" event and reduces stock.

5. Topic

A **Topic** is a category or storage location where messages are stored in Kafka.

Key Points:

- Topics act like folders for messages
- Producers write data into topics
- Consumers read data from topics
- Each topic has a unique name

Example Topics:

- order-events
 - payment-events
 - notification-events
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6. Broker

A **Broker** is a Kafka server that stores and manages data.

Responsibilities:

- Stores messages in topics
- Handles producer and consumer requests
- Manages message storage and retrieval

Key Point:

A Kafka cluster contains multiple brokers for scalability and fault tolerance.

7. Partition

A **Partition** is a subdivision of a topic.

Key Points:

- Each topic can have multiple partitions
- Partitions allow parallel processing
- Messages inside a partition are ordered

Example:

order-events topic:

- Partition 0
 - Partition 1
 - Partition 2
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8. Consumer Group

A **Consumer Group** is a group of consumers that work together to consume messages.

Key Points:

- Each message is processed only once within a group
- Enables load balancing
- Improves scalability

Example:

Topic: order-events

Consumer Group: inventory-group

Partition distribution:

- Partition 0 → Consumer 1
 - Partition 1 → Consumer 2
 - Partition 2 → Consumer 3
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9. Zookeeper (Optional in modern Kafka)

Zookeeper was used to manage Kafka brokers and cluster metadata.

Responsibilities:

- Broker coordination
- Leader election
- Cluster metadata storage

Note:

New Kafka versions can run without Zookeeper (KRaft mode).

10. Kafka Data Flow

Producer → Topic → Broker → Partition → Consumer Group → Consumer

11. Summary

- Producer sends messages to Kafka
 - Consumer reads messages from Kafka
 - Topic stores messages logically
 - Broker stores messages physically
 - Partition enables parallel processing
 - Consumer group enables scalability
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